ZetaERP — Manual Bug-Testing SOP

Part 6 — Tax (GST: registrations, rates, calculation, e-Invoice / e-Way Bill, GSTR-1)

Document	QA-SOP-06 — Tax
Product	ZetaERP / Zenith (multi-tenant POS / ERP)
Environment under test	Production — https://erp.zetaclantech.com
Scope of this part	The **Tax** module: Tax Dashboard, Tax Authorities, Tax Rates & GST Slabs, Tax Exemptions, **Tax Registrations (GSTIN)**, **GST Returns (GSTR-1)** — plus the **GST calculation engine** every billing screen depends on (intra- vs inter-state CGST/SGST/IGST, CESS, tax-inclusive pricing, composition, export/SEZ, reverse charge, TCS, rounding), **GSTIN validation**, **HSN/SAC**, **e-Invoice (IRN)** and **e-Way Bill**. (GSTR-2 / GSTR-3B reports live in **Accounting**, cross-referenced in §16. The stock-transfer IGST flow is Part 5.)
Audience	Manual QA tester (no code knowledge assumed; basic GST literacy helps)
Document owner	Platform / Solo-dev
Status	Active
Last updated	2026-07-01

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1. Purpose

This Standard Operating Procedure tells a tester **exactly how to manually test the Tax module end-to-end (A→Z)** — configuring tax authorities/rates/registrations, then proving the **GST calculation is legally correct** on every downstream document (POS bill, sales invoice, purchase invoice, stock transfer): the right **CGST+SGST vs IGST** split, the right **rate slab**, correct **CESS**, correct **tax-inclusive** extraction, correct **rounding**, correct **document type** (Tax Invoice / Bill of Supply / Export), and the compliance artefacts (**e-Invoice IRN**, **e-Way Bill**, **GSTR-1**).

> **Tax output is a legal and financial record.** A wrong CGST/SGST/IGST split, a wrong rate, or a wrong GSTIN on an issued invoice is a **compliance breach**, not a cosmetic bug. Do the math by hand for every calculation case. Never test on a live tenant's real invoices.

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2. Critical rules for the tester (read before starting)

> ⚠️ **This is a LIVE commercial system producing statutory GST documents.**

1. **NEVER create, edit, or file** a real tax registration, invoice, e-Invoice, e-Way Bill, or GSTR-1 return on the live flagship tenant **MARUTHI (org `100000`)** or any live tenant. Do all testing on a **throwaway tenant** you control.
2. **Never fabricate** real statutory identifiers — GSTIN, PAN, HSN/SAC, IRN, e-Way Bill numbers. Use clearly-fake, format-valid QA values (e.g. a dummy GSTIN with a valid state code and checksum only on a throwaway tenant). **Never** push a test document to the **real NIC e-Invoice / e-Way Bill portal** — confirm with the dev whether the environment is pointed at the NIC **sandbox** or a **mock**; if it's a real/production NIC credential, **do not generate**.
3. **Multi-tenant isolation is a key surface** (§15). Tax registrations, rates, exemptions, GSTINs and returns must be scoped to the active org. Any cross-tenant read/write of tax data is **Critical**.
4. **The GST engine is shared** by POS, Sales, Purchase and Stock Transfer. A change here affects all of them — verify the four callers still compute the same numbers.
5. **Clean up every test artefact** (registration, exemption, rate, invoice, filing, e-Invoice, e-Way Bill) — or hand the list to the dev. Do not file a return anywhere real.
6. **Show your working.** For every calculation case, record the inputs, the by-hand expected split, and the actual — a bare "looks right" is not acceptable for a GST case.

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3. Test environment & preconditions

Open from the app menu → **Tax**. Screen URLs (`/apps/zenith/tax/…` — lowercase):

Area	Screen (URL name)	Purpose
Dashboard	`TaxDashboard`	Tax module overview
Tax Authorities	`FindTaxAuthority` , `EditTaxAuthority?taxAuthorityId=`	Central/State GST authorities
Tax Rates	`FindTaxRate` , `EditTaxRate?taxRateId=`	Rates + read-only **GST Rate Slabs** list
Tax Exemptions	`FindTaxExemption` , `EditTaxExemption?exemptionId=`	Party/product exemptions (export, SEZ, RCM, etc.)
Tax Registrations	`FindTaxRegistration`	**GSTIN / PAN / TAN per party** (the intra/inter-state driver)
GST Returns	`Gstr1Returns?filingId=`	Generate / view / download **GSTR-1** (outward supplies)

Related (Accounting module, cross-referenced): GSTR-2 (`/apps/zenith/accounting/Gstr2Report`), GSTR-3B (`/apps/zenith/accounting/Gstr3bReport`), GST Reports (`/apps/zenith/accounting/GstReports`).

Engine (no screen, but tested through billing): `zenith.india.IndiaTaxServices.calculate#Tax` → `zenith.india.GstServices.calculate#Gst` . Configured per org via `CompanySettings.TAX_CALCULATION_SERVICE` .

Set up on the throwaway tenant: an **org GST config** (GSTIN + home state), a **customer with a GSTIN in the SAME state** (→ intra), a **customer with a GSTIN in a DIFFERENT state** (→ inter), an **unregistered (B2C) customer**, and **products across rate slabs** (0 / 5 / 18 / 28 / 28+CESS / exempt) with HSN codes. Have the **POS Terminal**, **Sales** (Create Invoice) and **Purchase** modules reachable to exercise the engine.

Browsers: full pass on Desktop Chrome; spot-check Edge/Safari/mobile. **Tools:** a **calculator** (mandatory — you verify GST math), DevTools (Console/Network), and the state-code table (01–38).

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4. Test data

Label	Use
Throwaway tenant	Non-MARUTHI org for all tax config + documents
Home state	The org's GST state, e.g. Karnataka `29`
`CUST_SAME`	Customer with a valid GSTIN in state **29** → intra-state (CGST+SGST)
`CUST_OTHER`	Customer with a valid GSTIN in a different state, e.g. Maharashtra `27` → inter-state (IGST)
`CUST_B2C`	Customer with **no GSTIN** (unregistered)
`PROD_18` / `PROD_5` / `PROD_28`	Products on the 18% / 5% / 28% slab, with HSN
`PROD_CESS`	Product on **28% + CESS** (e.g. `GST_28_CESS_12`)
`PROD_EXEMPT`	`isGstExempt = Y` (0%)
Fake GSTIN	Format-valid dummy per state, e.g. `29ABCDE1234F1Z5` (throwaway only)
Amounts to verify	qty 10 × ₹100 @18% → taxable 1000, tax 180 (intra: CGST 90 + SGST 90; inter: IGST 180)

State codes to know: `27` =Maharashtra, `29` =Karnataka, `33` =Tamil Nadu, `07` =Delhi (UT). Full range **01–38**.

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5. Conventions & severity

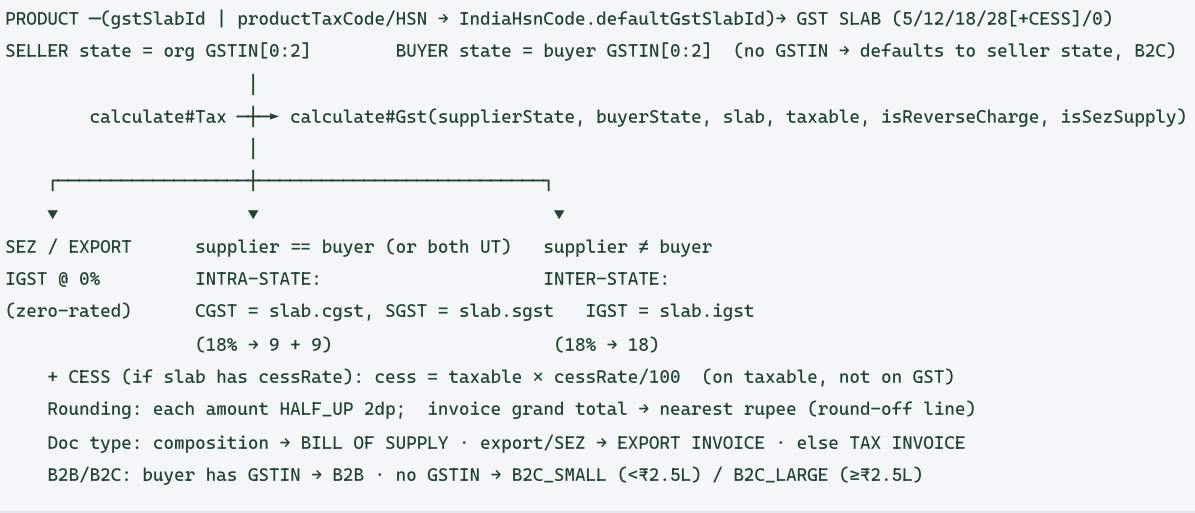
Case Type: `FUNC` · `CFG` (config) · `GST` (tax-math correctness) · `EDGE` · `UI` · `A11Y` · `SEC` · `ISO` · `PERF` .

Sev	Meaning	Example
S1 — Critical	Wrong GST math on a document; cross-tenant tax-data access; wrong GSTIN/state on an issued invoice; a real NIC push of test data	IGST charged on an intra-state sale; CGST+SGST charged on an inter-state sale; another tenant's GSTIN visible
S2 — High	A tax flow broken or a clearly wrong classification	engine returns zero tax for a taxed product; wrong document type (Tax Invoice vs Bill of Supply); GSTR-1 totals don't tie to invoices
S3 — Medium	Wrong/loose behaviour, limited impact	no GSTIN format check; rounding off by a paisa; exemption date range not honoured
S4 — Low	Cosmetic / minor UX	slab list mis-aligned; label typo; amount-in-words spacing

Logging a defect (template §17): Case ID, steps, **inputs + by-hand expected split vs actual** (mandatory for GST cases), tenant, seller/buyer state codes, browser, screenshot, Console/Network. Evidence is **mandatory** for GST, ISO and SEC.

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6. Flow map (how GST is decided)



7. TAX DASHBOARD

Screen: `TaxDashboard`.

ID	Type	Title	Steps	Expected result	Sev
TX-001	FUNC	Dashboard renders	Open the dashboard	Cards/metrics render; no console errors; org-scoped	S3
TX-002	FUNC	Quick links	Click each tile/link	Navigates to the right tax screen	S3
TX-003	UI	Empty state	New tenant with no tax config	No crash / NaN; sensible zero state	S3

8. CONFIG — Authorities, Rates & Slabs, Exemptions

8.1 Tax Authorities

Screens: `FindTaxAuthority`, `EditTaxAuthority`. Entity `TaxAuthority` (type National/State/..., country/region geo).

ID	Type	Title	Steps	Expected result	Sev
TX-010	CFG	List authorities	Open FindTaxAuthority	Lists authorities (e.g. Central GST, State GST); org-scoped where applicable	S3
TX-011	CFG	Create/edit authority	Create a State authority for a region → Save → edit	Persists; type + region correct	S3
TX-012	EDGE	Authority without region	Create a "State" authority with no region	Record actual — verify it's allowed/blocked sensibly	S3
TX-011a	SEC	**Authorities are platform-admin-only**	As a normal tenant, open FindTaxAuthority	**Read-only**: no "New", no Edit link, no Delete. Direct POST to create/update/delete → `Only a platform administrator ...`; nothing changes	S1
TX-011b	FUNC	Super-admin delete authority	As super-admin, delete an **unreferenced** authority vs a **referenced/parent** one	Unreferenced deletes; referenced → `Cannot delete: ... used by N record(s). Deactivate it instead.`	S2

8.2 Tax Rates & GST Slabs

Screen: `FindTaxRate` (+ read-only **GST Rate Slabs** list of `IndiaGstSlab`), `EditTaxRate`. Slabs carry `gstRate`, `cgstRate`, `sgstRate`, `igstRate`, `cessRate`.

ID	Type	Title	Steps	Expected result	Sev
TX-013	GST	Slab integrity	Open the GST Rate Slabs list	For every slab, **cgstRate + sgstRate == gstRate** and **igstRate == gstRate** (e.g. 18 → 9/9/18; 28 → 14/14/28; 5 → 2.5/2.5/5)	S1
TX-014	GST	CESS slabs	Inspect `GST_28_CESS_*` slabs	gstRate 28 (14/14/28) **plus** a separate `cessRate` (1/5/12/22)	S2
TX-015	GST	Exempt & special	Inspect `GST_EXEMPT`, precious-metal (0.1/0.25/1/3), composition (COMP_1/5/6)	Rates as labelled; exempt = 0	S3
TX-016	CFG	Create/edit a rate	Create a TaxRate under an authority → effective dates → Save	Persists; effective-from/thru honoured	S3
TX-017	EDGE	Overlapping effective dates	Two active rates for the same authority/category with overlapping dates	Record actual — which one wins; flag if ambiguous	S3
TX-016a	SEC	**Rates are platform-admin-only**	As a normal tenant, open FindTaxRate	Read-only (no New/Edit/Delete); the statutory GST slabs below stay visible. Direct POST → `Only a platform administrator ...`	S1

> **Note (ZE-TAX sweep):** Tax **Authorities & Rates are GLOBAL master data (no org)** — unlike Exemptions/Registrations they are NOT org-scoped, so every tenant sees the same list (correct). Create/edit/delete are restricted to super-admins via `store#` / `delete#TaxAuthority` and `store#` / `delete#TaxRate`; delete is refused while referenced. **Known pre-existing issue:** the zenith `tax_rate` FK points at mantle

> `tax_authority` (not `ZENITH_TAX_AUTHORITY`), so creating a zenith TaxRate against a zenith authority FK-fails — table is empty/unused (India uses `IndiaGstSlab`).

8.3 Tax Exemptions

Screen: `FindTaxExemption`, `EditTaxExemption`. Entity `TaxExemption` (party and/or product, type Full/ZeroRated/ReverseCharge/Export/SEZ/..., effective dates).

ID	Type	Title	Steps	Expected result	Sev
TX-018	CFG	Create party exemption	Exempt a party (e.g. Government) → Save	Created, org-scoped	S3
TX-019	FUNC	Exemption honoured	Bill a product/party with an **active** exemption	`check#TaxExemption` returns exempt → no GST charged on that line	S2
TX-020	EDGE	Expired exemption	Bill after the exemption's `effectiveThru`	Exemption **not** applied (GST charged); verify the date check	S3
TX-021	EDGE	Product vs party exemption	One line exempt-product, one taxable	Only the exempt line is zero-taxed	S2
TX-022	VAL	Mandatory fields on create	New Tax Exemption → leave Tax Authority / Exemption Type / both Party & Product blank → Create	**Blocked** with `Please fill the required field(s): …`; no record saved (ZE-TAX-002)	S2
TX-023	VAL	Mandatory fields on update	Open an exemption → clear Tax Authority / Type / Party+Product → Update	**Blocked** with the field list; **existing values unchanged** (not nulled)	S2
TX-024	EDGE	Effective date order	Set Effective Thru **before** Effective From → Save	Blocked: `Effective Thru cannot be before Effective From.`	S3
TX-025	UI	Edit & Delete actions present	Open the Tax Exemptions list	Each row shows an **Edit** link and a **Delete** action (ZE-TAX-004)	S2
TX-026	FUNC	Delete a record	Click **Delete** on a row → confirm	Row removed from list & DB; stays on the list	S2
TX-027	ISO	Cross-tenant delete blocked	Tenant B attempts to delete Tenant A's exemption (direct id)	**Denied** (`You do not have access to this tax exemption.`); row preserved	S1
TX-028	ISO	List is org-scoped	Log in as two tenants; compare the Exemptions list	Each tenant sees only its own exemptions; super-admin sees all	S1

> **Note (ZE-TAX-002/004):** at least **one of Party / Product** is required (an exemption may be party-based *or* product-based), plus Tax Authority and Exemption Type. Editing never wipes the record's org tag. Enforced server-side in `zenith.tax.TaxServices.store#TaxExemption` (create & update). **Delete** goes through `zenith.tax.TaxServices.delete#TaxExemption`, which verifies org ownership (ADMIN exempt) so a tenant can only delete its own records; the list find is org-scoped too.

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9. TAX REGISTRATIONS (GSTIN) — the intra/inter-state driver

Screen: `FindTaxRegistration`. Entities `TaxRegistration` (+ India `IndiaGstin`, `IndiaGstConfig`). The **buyer's and seller's `regionGeoId` (state)** — derived from their GSTIN — is what decides CGST+SGST vs IGST. `get#PartyTaxRegistration(partyId)` returns the default registration's `regionGeoId`.

ID	Type	Title	Steps	Expected result	Sev
TX-030	CFG	Org GST config	Set the org's GSTIN + home state (create#GstConfig / FindTaxRegistration)	GSTIN stored; `stateCode` = GSTIN[0:2]; a TrtGstin registration exists for the org	S1
TX-031	CFG	Register a customer's GSTIN	Add a GSTIN for `CUST_SAME` (state 29) and `CUST_OTHER` (state 27)	Each party's `regionGeoId` = its GSTIN state; `isDefault` set	S1

ID	Type	Title	Steps	Expected result	Sev
TX-032	GST	GSTIN → state extraction	Confirm the state code the system derives	`regionGeoId` matches GSTIN first two digits for every party	S1
TX-033	GST	**GSTIN format validation**	Enter an invalid GSTIN (too short; bad state code `99` ; wrong pattern; bad checksum)	`validate#Gstin` rejects with a clear reason; a valid 15-char GSTIN passes and returns state + PAN	S2
TX-033a	VAL	**GSTIN format on the Registrations screen**	New/Edit Registration → Type=GSTIN (India) → number `cfsadfsdgsd` → Save	**Rejected** (`That GSTIN doesn't look valid. A GSTIN is 15 characters…`); a valid 15-char GSTIN saves (normalised to upper-case); a lowercase valid GSTIN is upper-cased; non-GSTIN types (PAN/…) are not pattern-checked (ZE-TAX-007)	S2
TX-034	EDGE	Duplicate GSTIN	Register the **same** GSTIN twice	Blocked (unique index) — record actual	S2
TX-035	FUNC	Default vs multiple regs	Give a party two registrations, one `isDefault=Y`	`get#PartyTaxRegistration` returns the default (`-isDefault,-isPrimary` order)	S3
TX-036	EDGE	Party with no registration	A customer with **no** GSTIN/registration	`get#PartyTaxRegistration` returns nothing → downstream must handle (see TX-047) — record what the engine does	S2
TX-037	ISO	Registrations scoped	As tenant A, list registrations	Only A's parties/GSTINs; **no** tenant B rows (list find is org-filtered; super-admin sees all)	S1
TX-038	UI	Edit & Delete actions present	Open the Tax Registrations list	Each row shows an **Edit** link (→ EditTaxRegistration) and a **Delete** action (ZE-TAX-006)	S2
TX-039	VAL	Edit mandatory fields	Edit a registration → clear Party / Authority / Type / Number → Update	**Blocked** (`Please fill the required field(s): …`); row unchanged; reversed dates also blocked	S2
TX-040	FUNC	Delete a registration	Click **Delete** → confirm	Row removed from list & DB	S2
TX-041	ISO	Cross-tenant edit/delete blocked	Tenant B edits/deletes Tenant A's registration by direct id	**Denied** (`You do not have access to this tax registration.`); GSTIN preserved	S1

> **Note (ZE-TAX-006):** the Registrations list gained an Edit screen + Delete. **Edit** (`store#TaxRegistration`) and **Delete** (`delete#TaxRegistration`) both verify org ownership before mutating (ADMIN exempt) — critical because a registration's `regionGeoId` drives GST intra/inter-state, so cross-tenant tampering must be impossible. `get#PartyTaxRegistration` (the calc read) is unchanged. **ZE-TAX-007:** `create#TaxRegistration` + `store#TaxRegistration` now reject a malformed GSTIN when Type=GSTIN (India) and upper-case it (same regex as the Contacts path); onboarding callers use `ignore-error` , so an invalid GSTIN is simply skipped there rather than breaking signup.

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10. GST CALCULATION — correctness (the core; do the math)

Engine: `calculate#Tax` → `calculate#Gst` . Rate slab resolved per line: `taxCategoryId` → `productTaxCode` (HSN) → `product.gstSlabId` → `product.hsnCode` (HSN). Exercise it through **Sales → Create Invoice → Add Item**, the **POS Terminal**, and **Purchase**, and verify the numbers **by hand**. All amounts round **HALF_UP to 2 dp**.

10.1 Intra vs inter-state (the single most important thing)

ID	Type	Title	Inputs	Expected result (by hand)	Sev
GST-001	GST	Intra-state 18%	Seller 29, buyer 29 (`CUST_SAME`), qty 10 × ₹100, 18%	taxable **1000**; **CGST 90 + SGST 90**; IGST 0; line total **1180**	S1
GST-002	GST	Inter-state 18%	Seller 29, buyer 27 (`CUST_OTHER`), qty 10 × ₹100, 18%	taxable **1000**; **IGST 180**; CGST/SGST 0; line total **1180**	S1
GST-003	GST	5% slab intra	Seller 29, buyer 29, ₹1000, 5%	CGST 25 + SGST 25; total 1050	S1
GST-004	GST	28% slab inter	Seller 29, buyer 27, ₹1000, 28%	IGST 280; total 1280	S1

ID	Type	Title	Inputs	Expected result (by hand)	Sev
GST-005	GST	Both Union Territories	Seller UT, buyer same/other UT	Treated **intra-state** (CGST + **UTGST/SGST**), not IGST — verify the UT rule	S1
GST-006	GST	Mixed-rate invoice	Item A ₹1000 @5%, item B ₹2000 @18% (intra)	A: CGST 25/SGST 25; B: CGST 90/SGST 90; header **totalCgst 115, totalSgst 115**, taxable 3000	S1
GST-007	GST	Header totals aggregate	Any multi-line invoice	header totalCgst/totalSgst/totalIgst/totalCess = **sum of the lines**; taxable = sum of taxables	S1

10.2 CESS, inclusive pricing, rounding, exempt

ID	Type	Title	Inputs	Expected result (by hand)	Sev
GST-010	GST	CESS on top	₹1000, `GST_28_CESS_12`, intra	CGST 140 + SGST 140 + **CESS 120** (= 1000×12%); tax total 400; **CESS is on taxable, not on GST**	S2
GST-011	GST	Tax-inclusive (MRP)	MRP ₹118, 18%, `isTaxInclusive=Y`	taxable = 118×100/118 = **100.00**; tax **18.00** (`calculate#TaxInclusive`)	S1
GST-012	GST	Inclusive odd number	MRP ₹100, 18%, inclusive	taxable = 100×100/118 = **84.75** (HALF_UP); tax **15.25**; taxable+tax = 100.00	S2
GST-013	GST	Rounding per line	qty 3 × ₹99.99 @18% intra	taxable 299.97; CGST 26.9973→**26.997?** verify HALF_UP to 2dp (27.00) each side; no drift	S2
GST-014	GST	Invoice round-off	Grand total ₹1234.56	round-off line = round(1234.56) − 1234.56 = **+0.44**; payable **1235**	S2
GST-015	GST	Exempt product	`PROD_EXEMPT` (0%)	taxable recorded; **CGST/SGST/IGST/CESS all 0**; no tax lines	S2
GST-016	GST	0% vs exempt	A 0%-slab item vs an `isGstExempt` item	Both zero-tax; confirm doc-type impact (Bill of Supply for wholly-exempt supply)	S3

10.3 Document type, B2B/B2C, composition, export/SEZ, RCM, TCS

ID	Type	Title	Inputs	Expected result	Sev
GST-020	GST	B2B classification	Buyer **has** GSTIN	`b2bB2cType = B2B` ; documentType **Tax Invoice**	S2
GST-021	GST	B2C small vs large	Buyer no GSTIN, total ₹2,00,000 then ₹3,00,000	**B2C_SMALL** (<₹2.5L) then **B2C_LARGE** (≥₹2.5L)	S3
GST-022	GST	Composition → Bill of Supply	Org `isCompositionScheme=Y`	documentType **BILL OF SUPPLY** and **zero GST charged to the customer** (a composition dealer pays composition tax itself). **[Fixed b52c711 — `calculate#Tax` now returns zero tax for a composition org; verified tax 0, no CGST/SGST lines.]**	S1
GST-023	GST	Export invoice	Destination country ≠ IND (or `isExport=Y`)	documentType **EXPORT INVOICE**; **IGST 0%** under LUT (zero-rated). **[Fixed 809bc4e — `calculate#Tax` now passes the inferred `isExport` to the engine; exports default to 0% IGST (LUT), no CGST/SGST/CESS. Verified `EXPORT_LUT` → all 0. Override `isWithPayment=Y` → IGST at slab rate.]**	S1
GST-024	GST	SEZ supply	`isSezSupply=Y`	supplyType **SEZ**; **IGST 0%** under LUT (zero-rated) even intra-state. **[Fixed 809bc4e — SEZ now defaults to LUT 0% (was IGST at slab rate); `isWithPayment=Y` charges IGST for the with-payment model. Verified `SEZ_LUT` → all 0, `SEZ_WITH_IGST` → IGST 180.]**	S1
GST-025	GST	Reverse charge (RCM)	Invoice `isReverseCharge=Y`	Under RCM the **seller does not collect GST** (buyer self-assesses). **[Fixed b52c711 + wired 809bc4e — engine zeroes collected tax; the EditInvoice screen has a Reverse Charge toggle that flips the flag and re-taxes every draft line (verified: ON → item+header CGST/SGST 0; OFF → restored), and `add#InvoiceItem` / `create#SalesInvoice` pass the invoice's RCM flag.]** Verify `itcEligible` behaves per RCM	S1

ID	Type	Title	Inputs	Expected result	Sev
GST-026	GST	TCS threshold	Cumulative sales cross the threshold	TCS (0.1%) applies via `TcsCumulativeTracker` on the amount exceeding **₹50,00,000 (₹50 lakh) — confirmed correct per §206C(1H)** (code `limit=5000000`, rate 0.001). No change needed	S3

10.4 The engine's edge cases (probe hard)

ID	Type	Title	Inputs	Expected result	Sev
GST-030	EDGE	**B2C to another state**	`CUST_B2C` (no GSTIN) whose delivery is in a **different** state; goods sale	The engine defaults a **GSTIN-less** buyer to the **seller's** state → **intra-state (CGST+SGST)**. This is the correct legal default when the buyer's location is unknown (POS falls back to seller state). If the app later captures a B2C delivery state, place-of-supply should switch to IGST — record actual	S3
GST-031	EDGE	Consistent GSTIN-less default across paths	Same product billed via **POS**, **Sales** `add#InvoiceItem`, and the **Sales EditInvoice screen** with a GSTIN-less buyer	All three now default to **intra-state (CGST+SGST)**. **[Fixed b52c711 — the EditInvoice screen** `addItem` **previously defaulted** `isInterstate=true` **(wrongly IGST); now defaults intra, matching the engine and** `add#InvoiceItem` **.]**	S2
GST-032	EDGE	Unknown HSN	Product with an HSN not in `IndiaHsnCode`	No slab resolved → verify it errors clearly or falls back sensibly (not silently zero-tax)	S2
GST-033	EDGE	No slab on product	Product with no `gstSlabId` /HSN/category	Verify a clear failure, not a silent 0% (a taxed good billed at 0% is an under-charge)	S2
GST-034	EDGE	Negative / credit note	A credit note (negative amounts)	Tax reverses proportionally (negative CGST/SGST/IGST); split matches the original	S2
GST-035	PERF	Engine warm-up	First POS sale after opening a session	POS pre-warms the engine on `open#PosSession`; first real sale is fast (no ~1s cold-start)	S3

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11. HSN / SAC

Products carry `hsnCode` (8-char) / `sacCode`; `IndiaHsnCode` maps a code → `defaultGstSlabId`.

ID	Type	Title	Steps	Expected result	Sev
TX-050	FUNC	HSN → rate resolution	Product with only an HSN (no direct slab)	Engine resolves the slab from the HSN master	S2
TX-051	GST	HSN on the invoice/PDF	Print/inspect an invoice	Each line shows its **HSN/SAC**; GSTR-1 HSN summary groups by HSN + rate	S2
TX-052	EDGE	HSN validation	`validate#HsnSacCode` on a good vs junk code	Valid returns the default slab; junk is flagged	S3

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12. E-INVOICE (IRN)

Entities/services: `IndiaEinvoice`, `EinvoiceServices.generate#Einvoice/cancel#Einvoice`. Pushes a NIC-format JSON to the IRP and stores IRN + signed QR.

> ⚠️ **Confirm the target is NIC sandbox or a mock before generating.** Do not push test data to a real/production IRP.

ID	Type	Title	Steps	Expected result	Sev
TX-060	FUNC	Generate e-Invoice	On a B2B tax invoice, generate e-Invoice	Returns a 64-char **IRN**, ack no./date, signed QR; `IndiaEinvoice` row `GENERATED`. **[Fixed e744058 — was failing to persist (null PK); now works. Guards added: rejects Bill of Supply/composition, requires supplier GSTIN + a registered buyer (B2B) or export/SEZ + HSN on every line. Verified B2C correctly rejected.]**	S2

ID	Type	Title	Steps	Expected result	Sev
TX-061	GST	JSON payload correct	Inspect the NIC JSON	Real seller/buyer legal names (not placeholders), GSTIN + state, item HSN/qty/rate, **CGST/SGST vs IGST** matching the invoice, correct **SupTyp** (B2B/EXPWP/EXPWOP/SEZWP/SEZWOP) + **RegRev**, ValDtls incl. round-off tie to the invoice. **[Fixed e744058.]** ⚠ Still MOCK: the NIC API call (random IRN) — real filing needs NIC credentials	S1
TX-062	FUNC	Cancel e-Invoice	Cancel with a reason	Status `CANCELLED`, reason/date stored	S3
TX-063	EDGE	e-Invoice on B2C / exempt	Try to generate for a B2C or Bill-of-Supply doc	Handled per rule (e-Invoice is B2B/export only) — record actual	S3
TX-064	FUNC	QR code	Generate the QR	Base64 QR renders; encodes the signed invoice	S3

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13. E-WAY BILL

Entities/services: `IndiaEwayBill`, `EwayBillServices.generate#EwayBill/update#EwayBillPartB/cancel#EwayBill`. (Also see Part 5 — inter-state stock transfers.)

ID	Type	Title	Steps	Expected result	Sev
TX-070	FUNC	Generate e-Way Bill	For an invoice > ₹50,000, generate with transporter + vehicle	Returns an e-Way Bill number; `validUpto` = ceil(distance/200) days (min 1)	S2
TX-071	GST	Payload state codes	Inspect the payload	from/to GSTIN + state codes, HSN, taxable value, CGST value, transport mode correct	S2
TX-072	FUNC	Update Part B	Add/change the vehicle number	Part B updated	S3
TX-073	EDGE	Below threshold	Invoice ≤ ₹50,000	e-Way Bill not required — verify it's not forced	S3
TX-074	FUNC	Cancel	Cancel with a reason	Status updated	S3

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14. GST RETURNS — GSTR-1 (outward supplies)

Screen: `Gstr1Returns`. Services `generate#Gstr1Data`, `export#Gstr1Json`. Aggregates the period's sales invoices into B2B, B2C(S), and HSN summaries.

ID	Type	Title	Steps	Expected result	Sev
TX-080	FUNC	Generate GSTR-1	Pick a return period (MMYYYY) → Generate	A `Gstr1Filing` (DRAFT) with B2B / B2C(S) / HSN sections populated. **[Implemented e744058 — real aggregation replaced the mock stub; the screen's broken service call was repointed. Verified: B2B IGST 180, B2C(S) CGST/SGST 45, 2 HSN rows, real GSTN-schema JSON.]**	S2
TX-081	GST	**Totals tie to invoices**	Cross-check the return against the period's invoices	B2B taxable + tax = sum of B2B invoices; B2C(S) grouped by **place of supply + rate**; B2C(L) for inter-state unregistered >₹2.5L; HSN summary groups by HSN + rate; **grand totals reconcile**	S1
TX-082	GST	B2B vs B2C split	Mix registered + unregistered buyers	Registered → **B2B** (by buyer GSTIN); unregistered → **B2C(S)** (by POS + rate)	S1
TX-083	FUNC	Download JSON	Download the GSTR-1 JSON	Valid GSTN-schema JSON (gstin, fp, b2b[], b2cs[], hsn[])	S2
TX-084	EDGE	Cancelled invoices excluded	Include a cancelled invoice in the period	Excluded from the return	S2
TX-085	EDGE	Empty period	A period with no sales	Clean empty return; no crash	S3

15. MULTI-TENANT ISOLATION (high priority)

ID	Type	Title	Steps	Expected result	Sev
ISO-001	ISO	Registrations scoped	Tenant A lists Tax Registrations	Only A's parties/GSTINs	S1
ISO-002	ISO	Exemptions/rates/authorities scoped	A lists each config screen	Only A's rows (org-scoped ones); shared master data (slabs/states) is read-only reference	S1
ISO-003	ISO	GSTR-1 scoped	A generates/lists GSTR-1	Only A's invoices aggregated; B's never appear	S1
ISO-004	ISO	Direct-URL edit	A opens B's `EditTaxRate` / `EditTaxExemption` /registration by id	Blocked / not found	S1
ISO-005	ISO	e-Invoice / e-Way scoped	A views IRN/e-Way records	Only A's	S1
ISO-006	ISO	Engine uses the right org	Bill on tenant A	`calculate#Tax` uses A's GST config/state, never B's	S1

16. UI / ACCESSIBILITY / SECURITY (+ related reports)

ID	Type	Title	Check	Expected result	Sev
TX-090	UI	Desktop layout	All tax screens at 1920/1366	Slab/rate tables aligned; no overflow	S4
TX-091	UI	Mobile layout	Config + registration forms at 360–390px	No horizontal scroll; date pickers usable	S3
TX-092	A11Y	Keyboard	Tab through a registration/rate form	Logical order; Enter submits; visible focus	S3
TX-093	SEC	XSS in free text	`<img src=x onerror=alert(1)>` in legal name / exemption number / GSTIN label	Escaped everywhere (screens, PDFs, GSTR-1 JSON); never executes	S1
TX-094	SEC	CSRF	(Dev-assisted) POST a create-registration / generate-return without the session token	Rejected	S2
TX-095	FUNC	Related reports	Open Accounting → GST Reports / GSTR-2 / GSTR-3B	Render, org-scoped, tie to the ledger (deep-test in the Accounting SOP)	S3

17. Cross-flow / regression & defect template

ID	Type	Title	Expected result	Sev
XF-001	GST	Same product, four callers	Bill the same 18% item intra-state via **POS**, **Sales invoice**, **Purchase**, and inspect a **stock-transfer** IGST invoice → all compute CGST 9 + SGST 9 (POS/Sales/Purchase) and the transfer gives IGST inter-state; **numbers identical across callers**	S1
XF-002	GST	State change mid-flow	Change the buyer's GSTIN state between quote/order and invoice → split flips CGST/SGST ↔ IGST correctly	S1
XF-003	GST	Invoice → GSTR-1	An invoice's CGST/SGST/IGST + HSN flow into the correct GSTR-1 section and totals tie out	S1
XF-004	GST	Invoice → e-Invoice → e-Way	IRN JSON and e-Way payload carry the same GSTINs, states, HSN and tax split as the invoice	S1
XF-005	ISO	End-to-end isolation	Nothing tenant A configures or files is visible/usable by tenant B	S1

```
Defect ID:        ZE-TAX-____
Linked Case ID:   TX-0__ / GST-0__ / ISO-0__ / XF-0__
Title / Severity / Type:
Environment:      Prod | <browser> | tenant: <org>
Inputs:           seller state, buyer state, rate/slab, amount, inclusive?, RCM/SEZ/export flags
Expected (by hand): taxable, CGST, SGST, IGST, CESS, round-off, grand total, document type
Actual:           (same fields) - attach screenshot + Console/Network
Documents touched: invoice / registration / filing / IRN / e-Way ids
```

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18. Areas to probe hardest (attention map)

These are the highest-risk spots — spend extra time here:

1. **Intra vs inter-state split (GST-001/002, XF-001)** — the single most important thing. CGST+SGST for same-state, IGST for cross-state, driven by the parties' GSTIN state. Do the math by hand on every caller (POS, Sales, Purchase, Transfer).
2. **The default when a buyer has no GSTIN / no state (GST-030, GST-031, TX-036)** — all paths (POS, `add#InvoiceItem`, EditInvoice screen) now default a GSTIN-less buyer to the **seller's** state → intra-state (CGST+SGST), the correct default when the buyer location is unknown. **[Fixed b52c711 — the EditInvoice screen previously defaulted to IGST.]** Still re-verify the three callers agree after any tax change.
3. **Composition scheme (GST-022)** — must issue a **Bill of Supply with no GST charged to the customer**. **[Fixed b52c711 — `calculate#Tax` now returns zero tax for a composition org.]** Re-verify no CGST/SGST lines appear.
4. **Reverse charge (GST-025)** — the **seller must not collect** GST. **[Fixed b52c711 — `calculate#Gst` now zeroes the collected tax under `isReverseCharge=Y`.]** Re-verify the buyer isn't charged and ITC/accounting is right; the per-invoice RCM flag wiring from the UI is a separate follow-up.
5. **CESS (GST-010)** — charged on the **taxable value**, on top of GST, not on the GST amount.
6. **Tax-inclusive extraction & rounding (GST-011..014)** — `taxable = MRP×100/(100+rate)`, HALF_UP 2dp per amount, invoice total to nearest rupee. A drifting paisa on many lines compounds.
7. **Export / SEZ zero-rating (GST-023/024)** — 0% IGST, correct document type, LUT/shipping-bill fields.
8. **GSTR-1 reconciliation (TX-081/082, XF-003)** — the return's B2B/B2C(S)/HSN totals must tie exactly to the period's invoices; cancelled invoices excluded.
9. **e-Invoice / e-Way payload fidelity (TX-061/071, XF-004)** — GSTIN, state, HSN, and the CGST/SGST/IGST split in the JSON must match the invoice. **Never push test data to a real NIC portal.**
10. **Multi-tenant isolation of tax data (§15)** — GSTINs and returns are sensitive; any cross-tenant leak is Critical.

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19. Teardown (mandatory)

1. **List every test artefact** — registrations, exemptions, authorities/rates you added, invoices, GSTR-1 filings, e-Invoice (IRN) and e-Way Bill records — with ids.
2. **Do not file/submit** any return, e-Invoice or e-Way Bill to a real portal. If the environment was pointed at the real NIC by mistake, stop and tell the dev.
3. **Cancel/void** test invoices and their IRN/e-Way records; **delete** test registrations/exemptions/filings (or hand the list to the dev).
4. Delete any **exports/PDFs/JSON** generated on the test machine.
5. Confirm you **never touched MARUTHI** (org `100000`) or any live tenant's tax config or documents.
6. Attach the run sheet(s) + defect ids to this part's report.

End of Part 6 — Tax. Related: Part 3 — POS (billing calls the GST engine); Part 4 — Sales (invoice GST); Part 5 — Inventory (inter-state transfer IGST + e-Way); Accounting holds GSTR-2/GSTR-3B and the tax ledger.

Reviewed and fixed/confirmed: composition (zero GST / Bill of Supply), reverse-charge (seller collects zero + UI toggle), the GSTIN-less intra default, export/SEZ zero-rating (LUT 0% default with a per-invoice with-payment override), TCS (₹50 lakh @ 0.1%) — commits b52c711, 809bc4e; ***real GSTR-1 generation*** *(B2B/B2CL/B2CS/HSN/Export + GSTN JSON, replacing the mock),* ***e-Invoice & e-Way payload fidelity + compliance guards*** *(real party names, correct supply types, safe state/HSN, round-off, null-PK persistence), and* ***ITC treatment for composition/RCM vendor bills*** *— commits e744058, 48cc82a.*

The one remaining go-live dependency is the real NIC / GSTN portal integration *— the e-Invoice IRN, e-Way Bill number, and GSTR-1 filing are still generated locally as MOCKs (the JSON payloads are now faithful, but no data is transmitted). Wiring the live NIC/GSTN APIs needs the tenant's API credentials and is a deliberate go-live step — do not enable it against production without those credentials and a controlled cutover.*